

Homework 9

Exercises

1. Fix $a, b \in \mathbb{R}$ with $a < b$. Construct a bijection $[0, 1] \rightarrow [a, b]$.
2. In this exercise we will give an explicit bijection $\mathbb{Z} \rightarrow \mathbb{Q}$.
 - (a) Find a bijection $f : \mathbb{Z}^+ \rightarrow \mathbb{Z} \setminus \{0\}$.
 - (b) Consider $g : \mathbb{Z}^+ \rightarrow \mathbb{Q}^+$ defined as follows. For all $n \in \mathbb{Z}^+$ with $n > 2$, we know that n factors into primes: $n = p_1^{a_1} \cdots p_r^{a_r}$ where $1 < p_1 < \cdots < p_r$ and each $a_i \in \mathbb{Z}^+$. Then we define

$$g(n) = g(p_1^{a_1} \cdots p_r^{a_r}) := p_1^{f(a_1)} \cdots p_r^{f(a_r)},$$

where f is the function from part (a). Also define $g(1) = 1$. Prove that g is a bijection.

- (c) Use g to define a bijection $h : \mathbb{Z} \rightarrow \mathbb{Q}$.
3. Suppose $A \approx B$ and $C \approx D$. If $A \prec C$, show that $B \prec D$.
 4. If A is infinite and $B \subseteq A$ is finite, show that $A \setminus B$ must be infinite.
 5. If A is uncountable and $B \subseteq A$ is countable, show that $A \setminus B$ is uncountable.
 6. Let $\mathbb{Z}[x]$ be the set of polynomials with integers coefficients. Show that $\mathbb{Z}[x]$ is countable.
Hint: Can you write $\mathbb{Z}[x]$ as a countable union of countable sets?
 7. Construct an explicit bijection $(0, 1] \rightarrow (0, 1)$.
Hint: Try following the proof of the Cantor–Schröder–Bernstein Theorem.
 8. (Bonus) Some real (or even complex) numbers can be defined by an English phrase: e.g., “the ratio of the circumference of a circle to its diameter” defines the number π . We will consider the set of all such describable numbers. Give a loose argument (I can’t ask for a proof, since we don’t have formal definitions for any of the ideas, but **one can**) towards the following claims:
 - (a) The set Y of all grammatical sentences¹ of English is countably infinite.
Remember: Argue that Y is countable and that Y is infinite.
 - (b) The set Z of real numbers that can be defined by an English phrase is countably infinite.
 - (c) There are real numbers that cannot be defined by English phrases.
 9. (Bonus) Let $z \in \mathbb{C}$. Recall that $\overline{a + bi} = a - bi$. Prove that $\bar{z} = z$ if and only if $z \in \mathbb{R}$.
 10. (Bonus) Let $z \in \mathbb{C}$. Prove by induction that, for any $n \in \mathbb{Z}^+$, we have $\bar{z}^n = \overline{z^n}$.
 11. (Bonus) Let $z, w \in \mathbb{C}$ with polar forms $z = re^{\theta i}$ and $w = se^{\phi i}$ with $r > 0$. Fix $n \in \mathbb{Z}$. Prove z is invertible and show

$$z^{-1} = r^{-1}e^{-\theta i}, \quad z^n = r^n e^{\theta n i}, \quad \text{and} \quad \frac{w}{z} = \frac{s}{r} e^{(\phi - \theta)i}.$$

12. (Bonus) Compute all $z \in \mathbb{C}$ such that $z^3 = 1$, and draw them on the complex plane. Do the same for $z^4 = 1$ and $z^6 = 1$.
13. (Bonus) Fix $n \geq 2$. Show that the sum of the n th roots of unity (i.e., the solutions to $x^n = 1$) is 0.
14. (Bonus) Fix $n \geq 1$. What is the product of the n th roots of unity?

References

¹For simplicity, suppose that a grammatical sentence is a finite sequence of words.